Single Sine Excitation of a Nonlinear Activation Unit: Small Gain Signals

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Setup

```
clc; clear; close all;
SetDefaultPlotParameters;
```

Problem

A nonlinear activation unit with input gain γ is defined:

$$\varphi(v) = \tanh(\gamma v)$$

For input signal $v = \sin(\omega t + \phi)$ we wish to know the Fourier Series of the output signal $y = \varphi(v)$

That is we want to know the Fourier Series:

$$y(t) = \tanh(\gamma \sin(\omega t + \phi))$$

For now, we restrict out analysis to small gains $|\gamma| < \frac{\pi}{2}$.

Taylor Series Expansion

Let $x = \omega t + \phi$. Since $|\gamma| < \frac{\pi}{2}$ we may take the Taylor Series expansion of $\tanh(v)$:

$$y(x) = \sum_{m=1}^{\infty} a_m (\gamma \sin(x))^{2m-1}$$

$$y(x) = \sum_{m=1}^{\infty} a_m \gamma^{2m-1} \sin^{2m-1}(x)$$

Where:

$$a_m = \frac{2^{2m}(2^{2m} - 1) B_{2m}}{(2m)!}$$

Power Reduction Formula

We may apply the power reduction formula to reduce the $\sin^{2m-1}(x)$ factor to a trigonometric polynomial. The applicable power reduction formula is given via rule 1.320.3 on page 31 (PDF page 80) of Gradshteyn and Ryzik:

$$\sin^{2m-1}(x) = \frac{1}{2^{2m-2}}\sum_{k=0}^{m-1}(-1)^{m+k-1}C_k^{2m-1}\sin((2m-2k-1)x)$$

Thus:

$$y(x) = \sum_{m=1}^{\infty} a_m \gamma^{2m-1}\frac{1}{2^{2m-2}}\sum_{k=0}^{m-1}(-1)^{m+k-1}C_k^{2m-1}\sin((2m-2k-1)x)$$

$$y(x) = \sum_{m=1}^{\infty}\sum_{k=0}^{m-1} a_m \frac{(-1)^{m+k-1}\gamma^{2m-1}C_k^{2m-1}}{2^{2m-2}}\sin((2m-2k-1)x)$$

Substituting the definition of a_m:

$$y(x) = \sum_{m=1}^{\infty}\sum_{k=0}^{m-1}\frac{(-1)^{m+k-1}2^{2m}(2^{2m}-1)\gamma^{2m-1}C_k^{2m-1}B_{2m}}{2^{2m-2}(2m)!}\sin((2m-2k-1)x)$$

```
syms m
simplify(2^(2*m) / 2^(2*m-2))
```

```
ans =4
```

Simplifying we have:

%%%

```
%     2,1   3,1   4,1
%     3,2   4,2
%     4,3
%
```

Define a function yeilding the coefficient ξ associated with the m, k pair:

$$\xi(m,k) = \frac{(-1)^{m+k-1}4(2^{2m}-1)\gamma^{2m-1}C_k^{2m-1}B_{2m}}{(2m)!}$$

Now notice that due to the pattern above, we can evaluate the coefficient associated with each $\omega_n = 2n-1$

$$y(x) = \sum_{n=1}^{\infty} c_n \sin((2n-1)x)$$

Where:

$$c_n = \sum_{m=n}^{\infty} \xi(m, m-n)$$

Expanding:

$$y(x) = \sum_{n=1}^{\infty}\sum_{m=n}^{\infty} \frac{(-1)^{m+k-1}4(2^{2m}-1)\gamma^{2m-1}C_k^{2m-1}B_{2m}}{(2m)!}\sin((2n-1)x); k = m-n$$

Substituting $k = m-n$:

$$y(x) = \sum_{n=1}^{\infty}\sum_{m=n}^{\infty} \frac{(-1)^{m+(m-n)-1}4(2^{2m}-1)\gamma^{2m-1}C_{m-n}^{2m-1}B_{2m}}{(2m)!}\sin((2n-1)x)$$

$$y(x) = \sum_{n=1}^{\infty}\sum_{m=n}^{\infty} \frac{(-1)^{2m-n-1}4(2^{2m}-1)\gamma^{2m-1}C_{m-n}^{2m-1}B_{2m}}{(2m)!}\sin((2n-1)x)$$

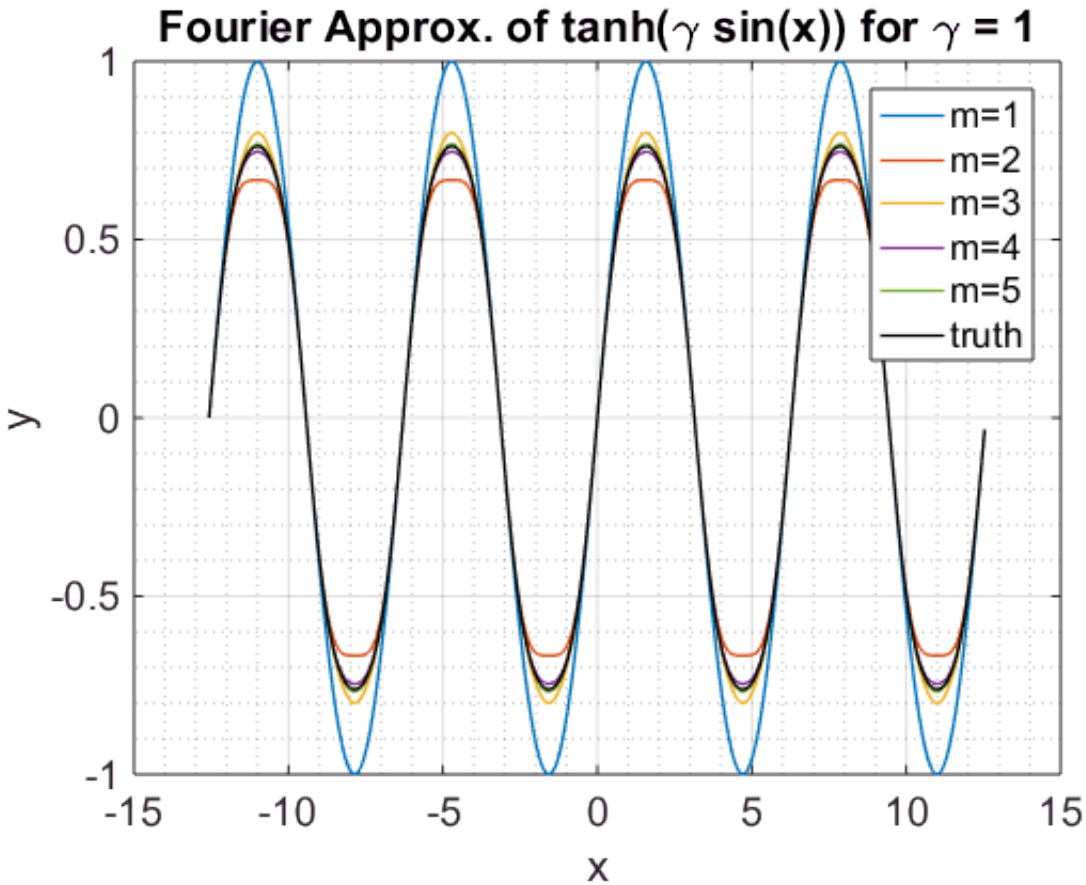
Thus we arrive at Theorem 2, which gives the sine series expansion of $\tanh(\gamma\sin(x))$

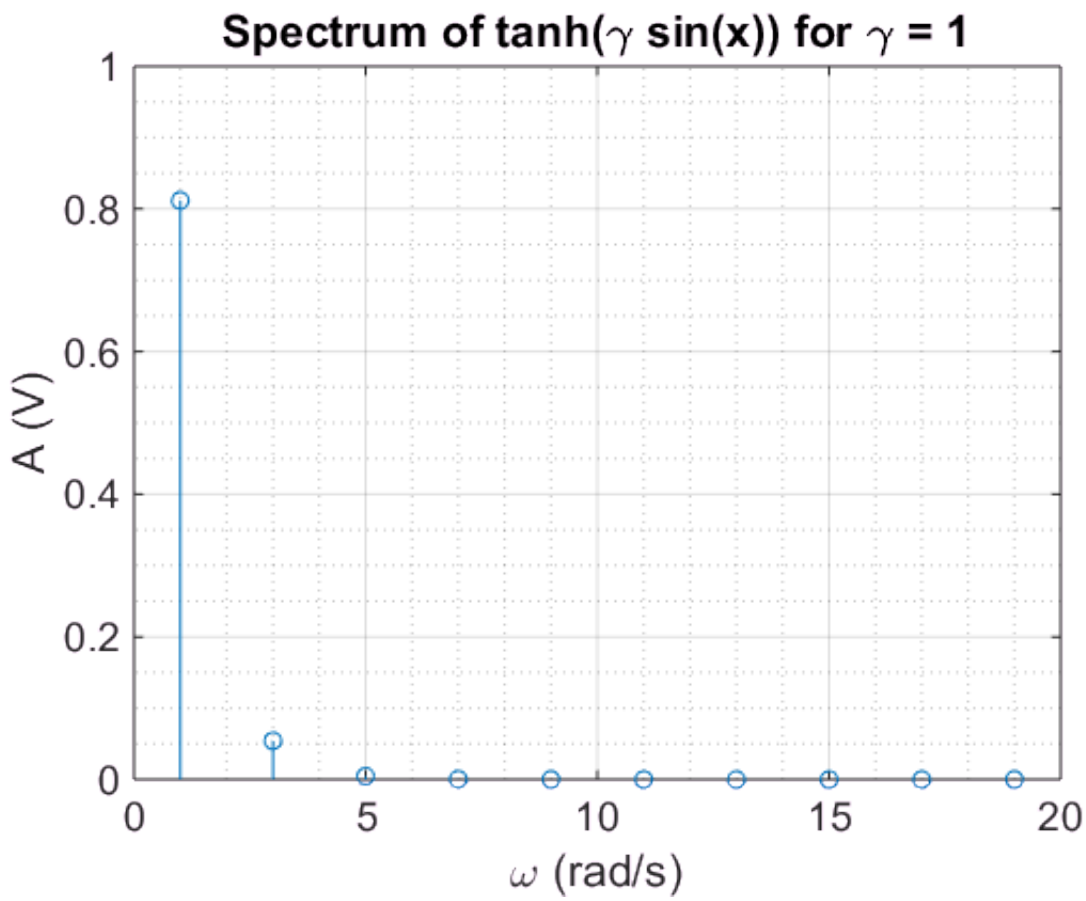
%%%

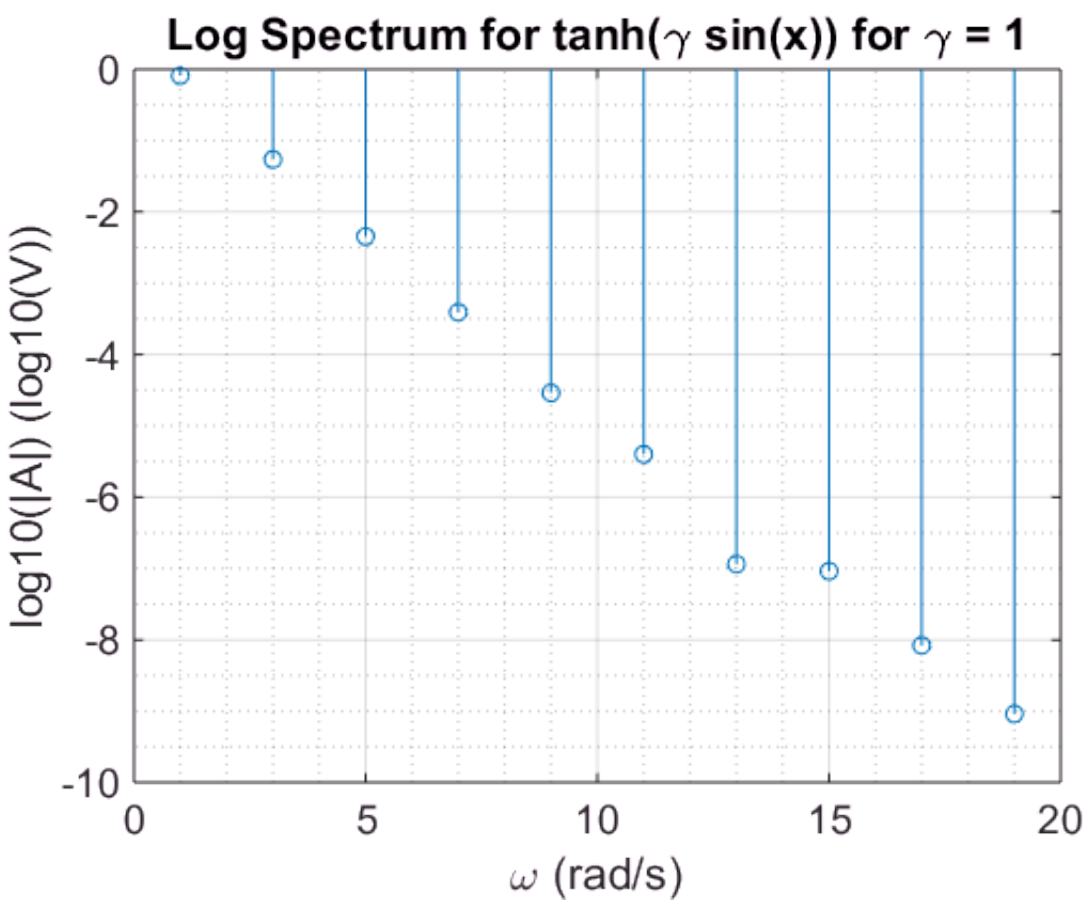
Testing the Result

Here we see the Fourier Series expansion of $y = \tanh(\gamma \sin(x))$ to $m = 1 \ldots 5$.

```
AnalyzeFourierSeries
```




Spectrum of tanh(γ sin(x)) for γ = 1
A (V)
ω (rad/s)
1
0.8
0.6
0.4
0.2
0
0
5
10
15
20


Log Spectrum for tanh(γ sin(x)) for γ = 1
log10(|A|) (log10(V))
ω (rad/s)
0
-2
-4
-6
-8
-10
0
5
10
15
20

Algorithms

```
function y = EvaluateFourierSineSeries(a, omega, x)
    y = zeros(size(x));
    for idx = 1:length(a)
        y = y + a(idx)*sin(omega(idx).*x);
    end
end

% @breif Analytic computation of the fourier series of tanh(gamma*sin(x))
% using Theorem 1.
function [a, omega] = GetFourierSeriesTanhSin1(model_order, gamma)
    a = [];
    omega = [];
    for m=1:model_order
        for k=0:m-1
            q_num = (-1)^(m+k-1)*4*(2^(2*m)-1)*gamma^(2*m-1);
            q_denom = (factorial(2*m));
            q = q_num/q_denom;
            c = nchoosek(2*m-1, k);
            b = bernoulli(2*m);

            a(end+1) = q*c*b;
            omega(end+1) = 2*m-2*k-1;
        end
    end
end

% @breif Analytic computation of the fourier series of tanh(gamma*sin(x))
% using Theorem 2.
function [a, omega] = GetFourierSeriesTanhSin2(model_order, gamma)
    a = [];
    omega = [];
    for n=1:model_order
        a_n = 0;
        for m=n:model_order
            q_num = (-1)^(2*m-n-1)*4*(2^(2*m)-1)*gamma^(2*m-1);
            q_denom = factorial(2*m);
            q = q_num/q_denom;
            c = nchoosek(2*m-1, m-n);
            b = bernoulli(2*m);
            a_n = a_n + q*c*b;
        end
        a(end+1) = a_n;
        omega(end+1) = 2*n-1;
    end
end


function AnalyzeFourierSeries
    gamma = 1;
    x = -4*pi:0.1:4*pi;
    y_truth = tanh(gamma*sin(x));
    legend_strs = {};

    for m = 1:10
        [a{m}, omega{m}] = GetFourierSeriesTanhSin2(m, gamma);
        y_sines{m} = EvaluateFourierSineSeries(a{m}, omega{m}, x);
```

```
    end

    figure;

    for m = 1:5
        plot(x, y_sines{m});
        hold on;
        legend_strs{end+1} = sprintf('m=%i', m);
    end

    legend_strs{end+1} = sprintf('truth');
    plot(x, y_truth, 'k');
    grid minor
    legend(legend_strs);
    xlabel('x');
    ylabel('y');
    title('Fourier Approx. of tanh(\gamma sin(x)) for \gamma = 1');

    figure;
    stem(omega{10}, a{10});
    hold on;
    grid minor
    xlabel('\omega (rad/s)');
    ylabel('A (V)');
    title('Spectrum of tanh(\gamma sin(x)) for \gamma = 1');

    figure;
    stem(omega{10}, log10(abs(a{10})));
    hold on;
    grid minor
    xlabel('\omega (rad/s)');
    ylabel('log10(|A|) (log10(V))');
    title('Log Spectrum for tanh(\gamma sin(x)) for \gamma = 1');

end


function SetDefaultPlotParameters
set(0, 'DefaultAxesFontSize', 14, ...
       'DefaultTextFontSize', 14, ...
       'DefaultAxesFontName', 'Arial', ...
       'DefaultTextFontName', 'Arial', ...
       'DefaultAxesXGrid','on', ...
       'DefaultAxesYGrid','on', ...
       'DefaultLineLineWidth', 1);
end
```